

Soal 6 optimasi Harga produk via Gradient Descent Manual

Parameter Personalisasi

$N = 10$	$b = 0$	$\theta_0 = 30^\circ$
$A = 1$	$k = 2$	$\alpha = 0.001$

Diketahui : model profit

$$\pi(x) = -(a+1) \cdot x^3 + (b+5) \cdot k \cdot x^2 + 100x - (k^2 \cdot 10), \quad x = \text{harga produk (Rp)}$$

6.1 Turunan dan titik kritis Manual

a. $\pi(x) = -(a+1) \cdot x^3 + (b+5) \cdot k \cdot x^2 + 100x - (k^2 \cdot 10)$

$$\pi(x) = -(1+1) \cdot x^3 + (0+5) \cdot 2 \cdot x^2 + 100x - (2^2 \cdot 10)$$

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

ditanya : $\pi'(x)$ menggunakan definisi limit.

$$\pi'(x) = \lim_{h \rightarrow 0} \frac{\pi(x+h) - \pi(x)}{h}$$

1. $\pi(x+h) = -2(x+h)^3 + 10(x+h)^2 + 100(x+h) - 40$

2. Ekspansi :

$$\begin{aligned}(x+h)^3 &= (x+h)(x+h)(x+h) \\ &= x^3 + 3x^2h + 3xh^2 + h^3\end{aligned}$$

$$\begin{aligned}(x+h)^2 &= (x+h)(x+h) \\ &= x^2 + 2xh + h^2\end{aligned}$$

3. substitusi ke fungsi

$$\begin{aligned}\pi(x+h) &= -2(x^3 + 3x^2h + 3xh^2 + h^3) + 10(x^2 + 2xh + h^2) + 100(x+h) - 40 \\ &= -2x^3 - 6x^2h - 6xh^2 - 2h^3 + 10x^2 + 20xh + 10h^2 + 100x + 100h - 40\end{aligned}$$

4. Kembali ke persamaan (Rumus diatas)

$$\pi(x+h) - \pi(x) :$$

$$\begin{aligned}&\Rightarrow (-2x^3 - 6x^2h - 6xh^2 - 2h^3 + 10x^2 + 20xh + 10h^2 + 100x + 100h - 40) - (-2x^3 + 10x^2 + 100x - 40) \\ &= \cancel{-2x^3} - 6x^2h - 6xh^2 - 2h^3 + \cancel{10x^2} + 20xh + 10h^2 + \cancel{100x} + 100h - \cancel{40} + \cancel{2x^3} - \cancel{10x^2} - \cancel{100x} + \cancel{40} \\ &= -6x^2h - 6xh^2 - 2h^3 + 20xh + 10h^2 + 100h \\ &= h(-6x^2 - 6xh - 2h^2 + 20x + 10h + 100)\end{aligned}$$

5) masukkan ke Rumus

$$\pi'(x) = \lim_{h \rightarrow 0} \frac{h(-6x^2 - 6xh - 2h^2 + 20x + 10h + 100)}{h}$$

$$\lim_{h \rightarrow 0} (-6x^2 - 6xh - 2h^2 + 20x + 10h + 100)$$

6) substitusi $h=0$

$$\pi'(x) = -6x^2 - 6x(0) - 2(0)^2 + 20x + 10(0) + 100$$

$$\pi'(x) = -6x^2 + 20x + 100$$

Hasil Akhir

$$\pi'(x) = -6x^2 + 20x + 100$$

b) $\pi''(x)$ & $\pi'''(x)$ dengan aturan pangkat

Jawaban

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

$$\pi'(x) = -6x^2 + 20x + 100$$

rumus aturan pangkat

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\pi''(x) = -2(-6x^1) + 1(20) = 12x + 20$$

$$\pi'''(x) = 1(12) + 0 = 12$$

Tabel Hasil Turunan

Orde Turunan	Bentuk Fungsi
$\pi(x)$	$-2x^3 + 10x^2 + 100x - 40$
$\pi'(x)$	$-6x^2 + 20x + 100$
$\pi''(x)$	$12x + 20$
$\pi'''(x)$	12

c) Titik kritis dan uji maksimum/minimum.

Diketahui :

$$\pi'(x) = -6x^2 + 20x + 100$$

$$\pi''(x) = 12$$

1) Titik kritis

$$\text{syarat titik kritis } \pi'(x) = 0$$

$$\pi'(x) = 0$$

Rumus kuadrat

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

Rumus Diskriminan

$$D = b^2 - 4ac$$

Ketentuan maksimum dan minimum (menguji turunan kedua)

Jika: $\pi''(x) > 0$ minimum lokal

$\pi''(x) < 0$ maksimum lokal.

Penyelesaian

$$\Rightarrow \pi'(x) = 0$$

$$-6x^2 + 20x + 100 = 0$$

$$a = -6, b = 20, c = 100$$

D, Diskriminan

$$D = b^2 - 4ac$$

$$D = 20^2 - 4(-6)(100)$$

$$D = 400 + 2400$$

$$D = 2800$$

D, masukkan ke rumus kuadrat

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

$$= \frac{-20 \pm \sqrt{2800}}{2(-6)}$$

$$= \frac{-20 \pm 10\sqrt{28}}{-12}$$

$$= \frac{-20 \pm 20\sqrt{7}}{-12}$$

$$= \frac{-20(-1 \pm \sqrt{7})}{-12}$$

$$= \frac{5}{3}(-1 \pm \sqrt{7})$$

$$x_1 \text{ (positif)} = \frac{-5}{3}(-1 + \sqrt{7})$$

$$x_2 \text{ (negatif)} = \frac{-5}{3}(-1 - \sqrt{7})$$

$$x_1 = \frac{5}{3}(1 - \sqrt{7})$$

$$x_2 = \frac{5}{3}(1 + \sqrt{7})$$

Titik Akhir

$$x_1 = \frac{5}{3}(1 - \sqrt{7})$$

$$x_2 = \frac{5}{3}(1 + \sqrt{7})$$

2) Minimum / maximum lokal

uji turunan ke-2

$$\pi''(x) = -12x + 20$$

uji di x_1

$$\pi''(x) = -12 \left(\frac{5}{3} (1 - \sqrt{7}) \right) + 20$$

$$= -4 (5 - 5\sqrt{7}) + 20$$

$$= -20 + 20\sqrt{7} + 20$$

$$= 20\sqrt{7}$$

$$\pi''(x) = 20\sqrt{7} > 0 \quad (\text{minimum})$$

uji di x_2

$$\pi''(x) = -12 \left(\frac{5}{3} (1 + \sqrt{7}) \right) + 20$$

$$= -20 (1 + \sqrt{7}) + 20$$

$$= -20 - 20\sqrt{7} + 20$$

$$= -20\sqrt{7}$$

$$\pi''(x) = -20\sqrt{7} < 0 \quad (\text{maksimum})$$

• titik $x_1 = \frac{5}{3} (1 - \sqrt{7}) \rightarrow$ minimum lokal

titik $x_2 = \frac{5}{3} (1 + \sqrt{7}) \rightarrow$ maksimum lokal

6.2 Gradient Descent Manual

• minimalkan profit $-\pi(x)$ menggunakan Gradient descent

• $x_0 = 1$ • awal

• $\alpha = 0,001$

• Update Rule Gradient

$$x_{k+1} = x_k + \alpha \cdot \pi'(x_k)$$

$$k_{k+1} = k - \alpha \cdot \pi'(x_k)$$

d) 5 iterasi Gradient Descent dan tabel

Di iterasi 0 ($k=0$), iterasi pertama

a) nilai x_0

$$x_0 = 1. (6 \text{ angka di belakang koma})$$

$$x_0 = 1.000.000$$

b) $\pi(x_0) \rightarrow x_0 = 1$

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

$$\pi(1) = -2(1)^3 + 10(1)^2 + 100(1) - 40$$

$$= -2 + 10 + 100 - 40$$

$$= 68.000.000$$

c) $\pi'(x_0)$

$$\pi'(x) = -6x^2 + 20x + 100$$

$$\pi'(1) = -6(1)^2 + 20(1) + 100$$

$$\pi'(1) = -6 + 20 + 100$$

$$\pi'(1) = 114.000.000$$

$$d) x_{k+1} = x_k + \alpha \pi'(x_k)$$

$$x_{0+1} = x_0 + \alpha \cdot \pi'(x_0)$$

$$x(1) = 1 + (0,001 \times 114)$$

$$= 1 + 0,114$$

$$= 1.114.000$$

➤ Iterasi 1 ($k=1$), Iterasi ke-2

a) $x_1 = 1.114.000$

b) $\pi(x_1)$

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

$$x_1 = 1.114$$

$$x_1^2 = 1.241.796$$

$$x_1^3 = 1.383.061$$

$$\pi(x_1) = -2(1.383.061) + 10(1.241.796) + 100(1.114) - 40$$

$$\begin{aligned}\pi(1.114) &= -2.767.022 + 12.417.960 + 111.400.000 - 40 \\ &= 01.050.038\end{aligned}$$

c) $\pi'(x_1)$

$$\pi'(x) = -6x^2 + 20x + 100$$

$$\pi'(1.114) = -6(1.241.796) + 20(1.114) + 100$$

$$= -7.450.776 + 22.280.000 + 100$$

$$= 114.829.224$$

d) $x_{(k+1)} = x_{(k)} + \alpha \cdot \pi'(x_1)$

$$x_{(2)} = 1.114 + 0.001 \cdot 114.829.224$$

$$= 1114 + 114.829,224$$

$$x_2 = 1.228.829$$

➤ Iterasi 2 ($k=2$), Iterasi ke-3

a) $x_2 = 1.228.829$

$$x_2^2 = 1.510.031$$

$$x_2^3 = 1.855.004$$

b) $\pi(x_2)$

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

$$= -2(1.855.004) + 10(1.510.031) + 100(1.228.829) - 40$$

$$= -3.710.008 + 15.100.310 + 122.882.900 - 40$$

$$= 94.273.202$$

$$\pi(x_2) = 94.273.202$$

c) $\pi'(x_2)$

$$\pi'(x) = -6x^2 + 20x + 100$$

$$= -6(1.510.031) + 20(1.228.829) + 100$$

$$= -9.060.186 + 24.576.580 + 100$$

$$= 115.516.394$$

$$\begin{aligned} d) \quad x_{(2+1)} &= x_2 + \alpha \cdot \pi'(x_2) \\ x_3 &= 1.288826 + (0.001 \cdot 115.516394) \\ &= 1.344345 \end{aligned}$$

⇒ Iterasi 3 ($k=3$), Iterasi ke-4

$$\begin{aligned} a) \quad x_3 &= 1.344345 \\ x_3^2 &= 1.807268 \\ x_3^3 &= 2.429928 \end{aligned}$$

$$b) \quad \pi(x_3)$$

$$\begin{aligned} \pi(x) &= -2x^3 + 10x^2 + 100x - 40 \\ \pi(x_3) &= -2(2.429928) + 10(1.807268) + 100(1.344345) - 40 \\ &= 107.647394 \end{aligned}$$

$$c) \quad \pi'(x_3)$$

$$\begin{aligned} \pi'(x) &= -6x^2 + 20x + 100 \\ &= -6(1.807268) + 20(1.344345) + 100 \\ &= 116.043292 \end{aligned}$$

$$d) \quad x_4$$

$$\begin{aligned} x_{(3+1)} &= x_3 + \alpha \cdot \pi'(x_3) \\ x_4 &= 1.344345 + (0.001 \cdot 116.043292) \\ &= 1.460388 \end{aligned}$$

⇒ Iterasi 4 ($k=4$), Iterasi ke-5

$$\begin{aligned} a) \quad x_4 &= 1.460388 \\ x_4^2 &= 2.132734 \\ x_4^3 &= 3.113431 \end{aligned}$$

$$b) \quad \pi(x_4)$$

$$\begin{aligned} \pi(x) &= -2x^3 + 10x^2 + 100x - 40 \\ \pi(x_4) &= -2(3.113431) + 10(2.132734) + 100(1.460388) - 40 \\ &= -6.226862 + 21.327340 + 146.038800 - 40 \\ &= 121.139278 \end{aligned}$$

$$\pi(x_4) = 121.139278$$

$$c) \quad \pi'(x_4)$$

$$\begin{aligned} \pi'(x) &= -6x^2 + 20x + 100 \\ \pi'(x_4) &= -6(2.132734) + 20(1.460388) + 100 \\ &= -12.796404 + 29.207760 + 100 \\ &= 116.411356 \end{aligned}$$

$$d) \quad x_5$$

$$\begin{aligned} x_{(4+1)} &= x_4 + \alpha \cdot \pi'(x_4) \\ x_5 &= 1.460388 + 0.001 \cdot 116.411356 \\ &= 1.576799 \end{aligned}$$

Tabel Hasil Iterasi

k	x_k	$\pi(x_k)$	$\pi'(x_k)$	x_{k+1}
0	1.000000	68.000000	114.000000	1.114000
1	1.114000	81.050038	114.826224	1.228829
2	1.228829	94.273202	115.516394	1.344345
3	1.344345	107.627324	116.043292	1.460388
4	1.460388	121.139228	116.411356	1.576789

(e) Pengerjaan di matlab.

(f) Pengerjaan di matlab.

6.3 Aplikasi Bisnis

(g) Diketahui :

Tilut maksimum lotus (dari soal 6.1 c)

$$x^* = \frac{5}{3} (1 + \sqrt{7})$$

Fungsi profit

$$\pi(x) = -2x^3 + 10x^2 + 100x - 40$$

$$K = 2$$

$$\text{Harga bahan baku/unit} = (k \cdot 10) \text{ ribu} = 20.$$

Ditanya :

a) konversi x^* ke harga rupiah

b) profit maksimum $\pi(x^*)$

c) margin/unit

Penglesaian

1) Hitung x^*

$$\begin{aligned} x^* &= \frac{5}{3} (1 + \sqrt{7}) \\ &= \frac{5}{3} (1 + 2.645751) \\ &= \frac{5}{3} (3.645751) \end{aligned}$$

$$= 6.076252$$

2) konversi ke rupiah (dalam ribuan)

$$\text{Harga optimal} = 6.076 //$$

b/ Profit maksimum

$$\pi(x^*) = -2x^3 + 10x^2 + 100x - 40$$

$$(x^*)^2 = 36,981$$

$$(x^*)^3 = 224,360$$

$$\begin{aligned}\pi(x^*) &= -2(224,360) + 10(36,981) + 100(6,076252) - 40 \\ &= -448,720 + 369,810 + 607,625 - 40 \\ &= 488,115\end{aligned}$$

Dalam Rupiah

$$\pi(x^*) = \text{Rp. } 488.115 //$$

b/ Margin

$$\text{margin} = \text{Harga} - \text{Biaya.}$$

$$= 6,076252 - 20$$

$$= -13,923748 //$$

(margin negatif artinya harga jual lebih rendah daripada biaya)

h)

Gradient Descent bisa terjebak di local minimum jika fungsi yang dioptimasi memiliki banyak titik minimum atau bentuknya tidak sederhana. Local minimum adalah titik terendah sementara, bukan yang paling optimal secara keseluruhan. Hal ini biasanya terjadi ketika pemilihan titik awal x_0 yang kurang tepat atau bentuk kurva yang memiliki banyak lembah. Dalam soal 909, fungsi profit relatif sederhana sehingga kemungkinan terjebak kecil, tetapi pada kasus nyata seperti machine learning, fungsi lebih kompleks dan kita saja memiliki banyak titik kritis. Untuk menghindarinya dapat digunakan beberapa strategi seperti mencoba beberapa titik awal berbeda, atau memilih learning rate yang tepat.

Contoh nyata: training model neural network machine learning, model bisa stuck atau terjebak di local minimum sehingga akurasi-nya tidak maksimal.